DESIGN NOTE · WEEK 42

How the orders service serves a read

A read of one order is answered from the cache when it can be, and from **Postgres** when it cannot.

READS A DAY	CACHE HITS	P99
612 000	91 %	166 ms
+8 % on week 41	down from 94 %	under the 200 ms budget

The path

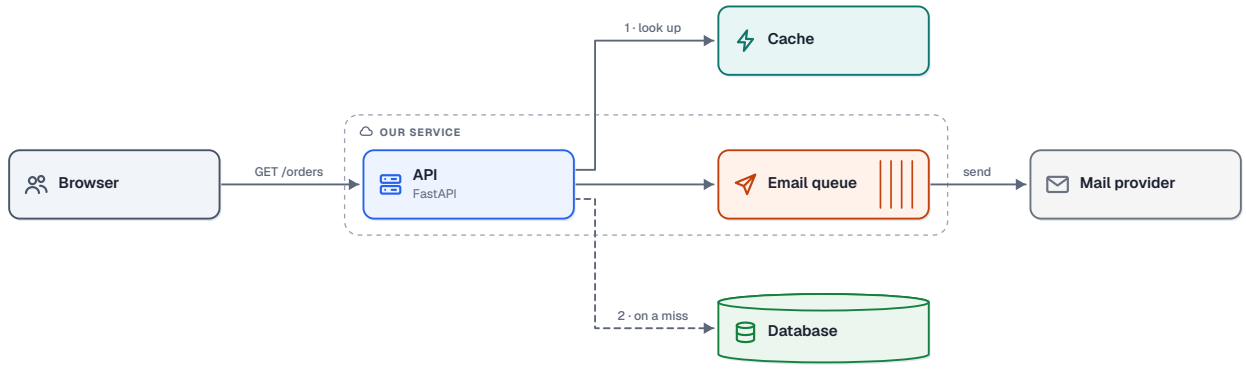


Figure 1: The API asks the cache first and the database only on a miss (the dashed arrow).

> **Key idea**
> A miss is not an error, only a slower read; the API fills the cache on its way back.

One request, message by message

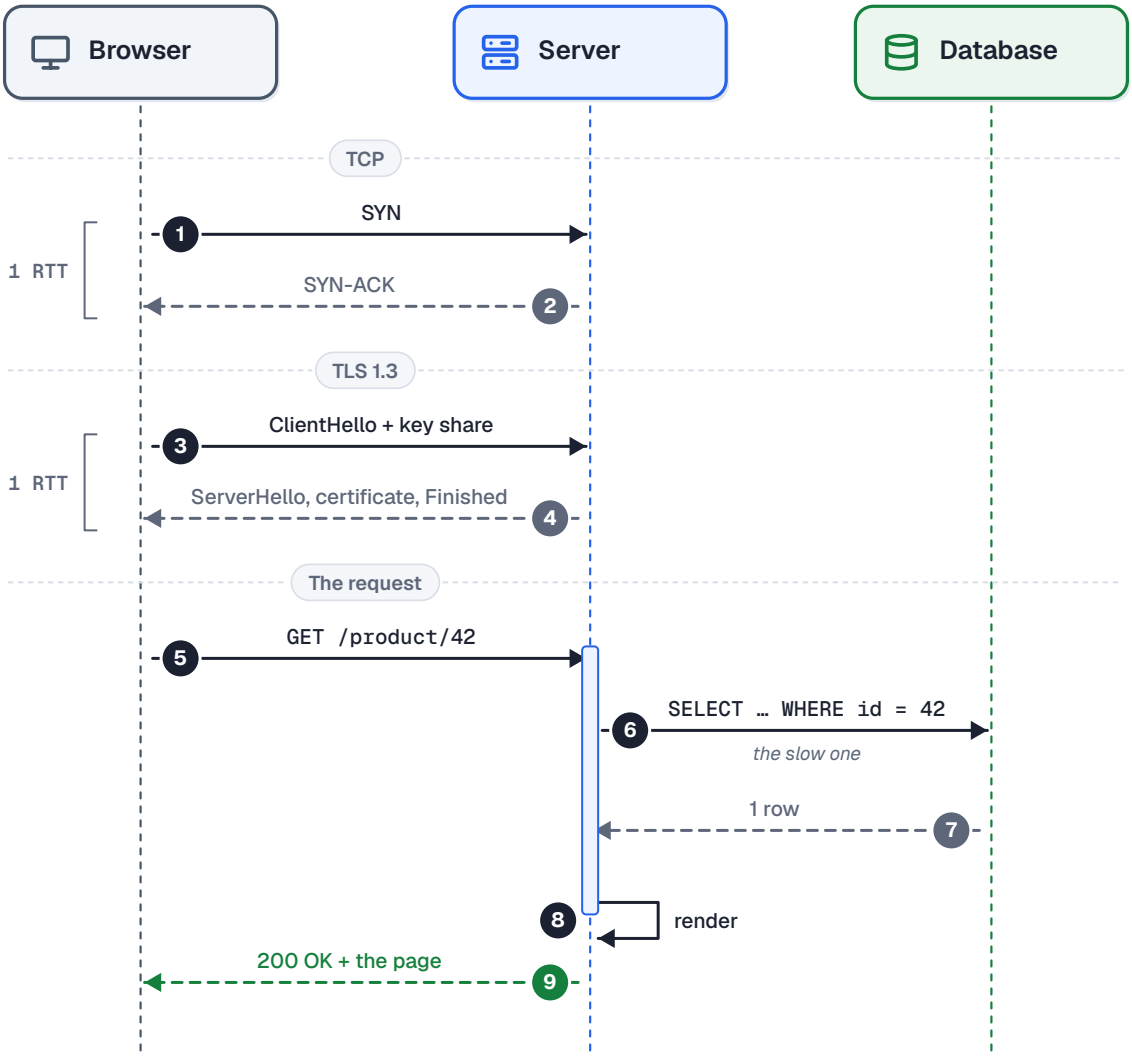


Figure 2: Two round trips go to TCP and TLS before the request itself is sent.

What it costs

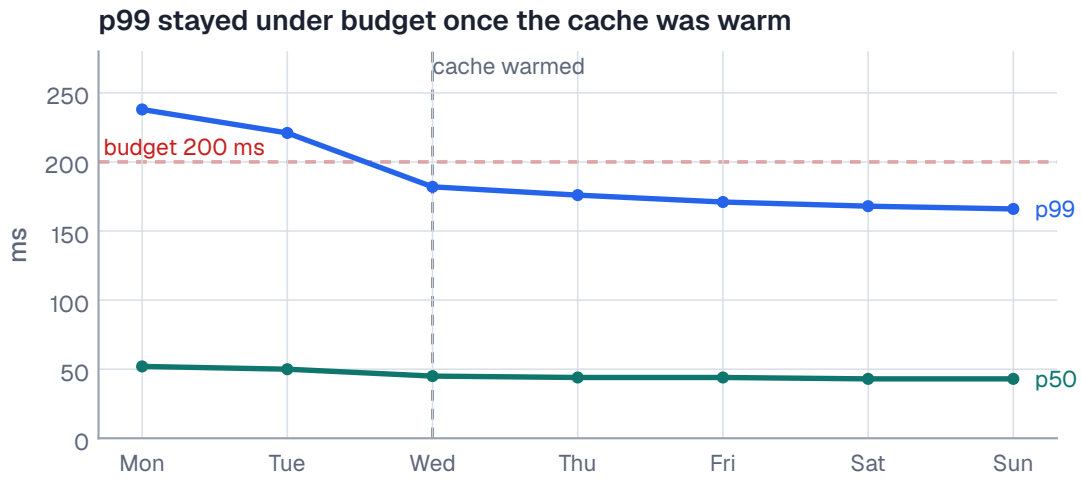


Figure 3: p99 fell under its budget on Wednesday, when the cache had warmed.

Endpoint	Reads a day	p50	p99
GET /orders/{id}	612 000	44 ms	212 ms
GET /orders?customer=	148 000	61 ms	180 ms

Table 1: The two busiest reads last week.

```
def read(order_id):
    hit = cache.get(order_id)
    return hit if hit else fill(order_id)
```

Listing 1: The read, in three lines.